

HAT-P-12b

R-0807

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_220324 · created 2026-07-28T11:17:48+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459659.6915 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.187 +/- 0.014
Transit depth (Rp/Rs)^2	3.52 +/- 0.53 [%]
Orbital Inclination (inc)	89.79 +/- 0.68
Airmass coefficient 1 (a1)	1.885 +/- 0.026
Airmass coefficient 2 (a2)	-0.59 +/- 0.16
Scatter in the residuals of the lightcurve fit is	1.38 %
Best Comparison Star	#3 - [225, 161]
Optimal Method	PSF photometry
Transit Duration (day)	0.035 +/- 0.0024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.187 ± 0.014 vs 0.1406 (deviation 2.8σ)
Duration fit vs published	0.035 d vs 0.0975 d (deviation 21.98σ)
Mid-time O-C	-4629.8 min
Depth SNR	11.3
Residual scatter	1.38 %

Light curve

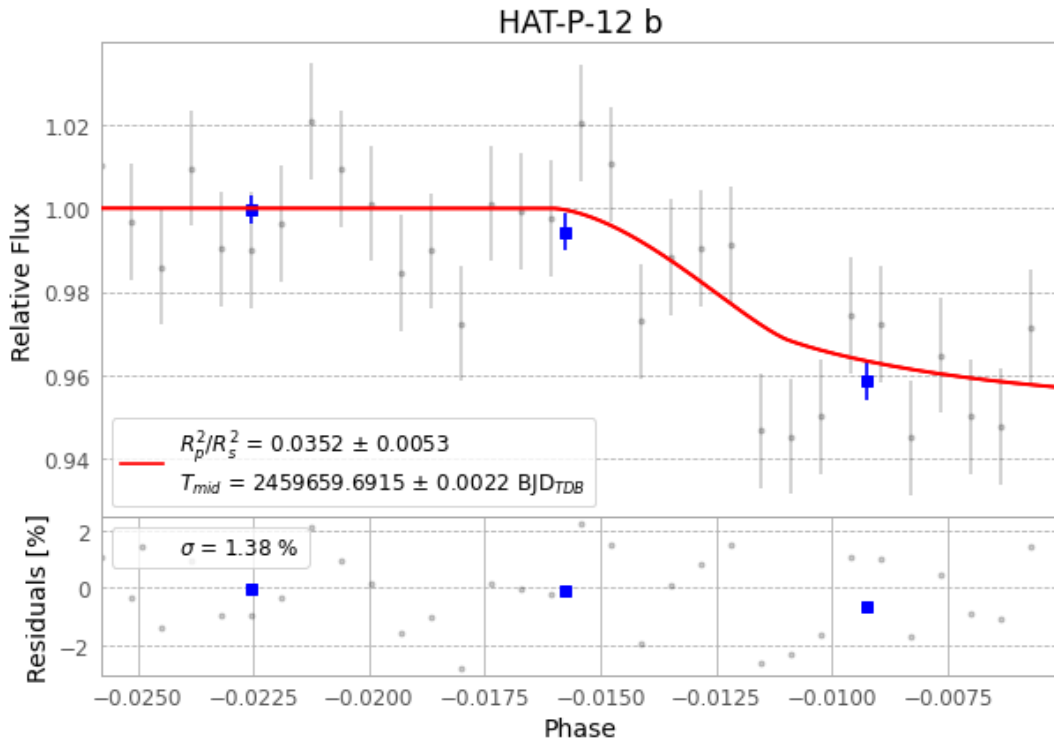


Figure 1 — FinalLightCurve_HAT-P-12 b_2022-03-24.png (SHA-256 cce11329c49b1ed4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [201, 233], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2f63d681ea22f4d6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

33 FITS frames (22 MB), HATP-12220324073615.FITS → HATP-12220324091212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-220324.log (a635f55f0e75...), FinalLightCurve_HAT-P-12 b_2022-03-24.png (cce11329c49b...), FinalLightCurve_HAT-P-12 b_2022-03-24.csv (7312673cfc6f...)

Compute resources

Wall time 85.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 11:17Z.